h(t) tide at the mouth at time *t* (m above mean sea level)

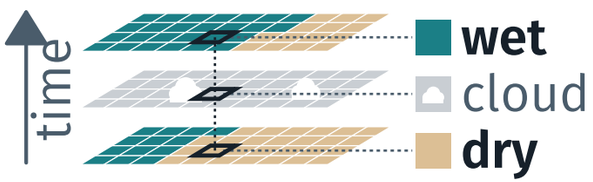
τ delay behind the mouth (min; fitted per 3 km of channel)

z ground elevation of the pixel (m, same scale as *h*)

INPUT 1 · IMAGES

1 Seen many times

Sentinel-2 data cube



10-m pixels · 2023–2025

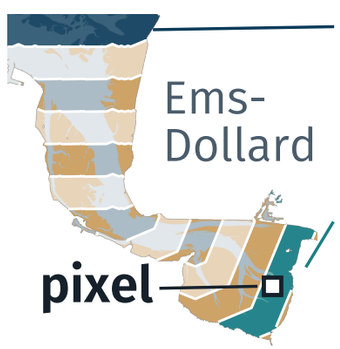
each look: wet or dry

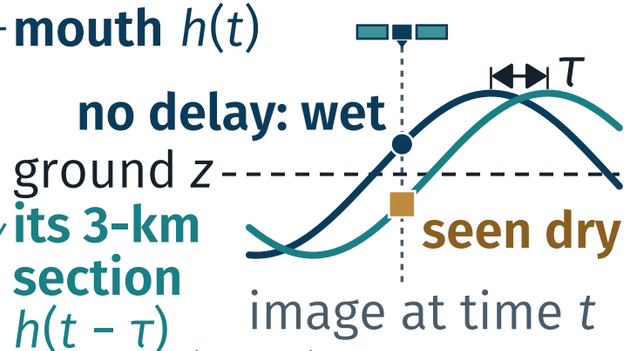
1 look = 1 cloud-free view

water index from green and near-infrared light

INPUT 2 · TIDE AT THE MOUTH

2 Its tide arrives late





wet when *h*(*t* − *τ*) > *z*

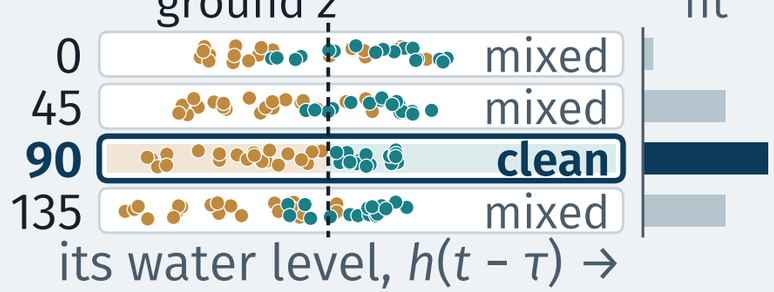
the mouth tide reaches its section *τ* later

MAREA shifts the tide in time, not height

3 Its section's delay

delays *τ* tried (minutes, examples)

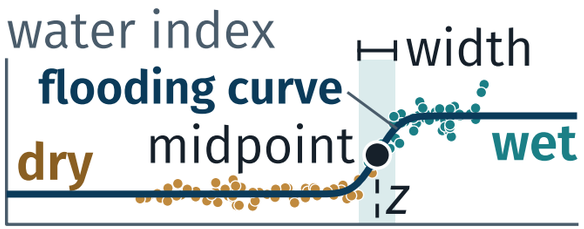
dry **wet**: one look of this pixel



best delay: cleanest split

fit: judged on its section's pixels

4 Its elevation

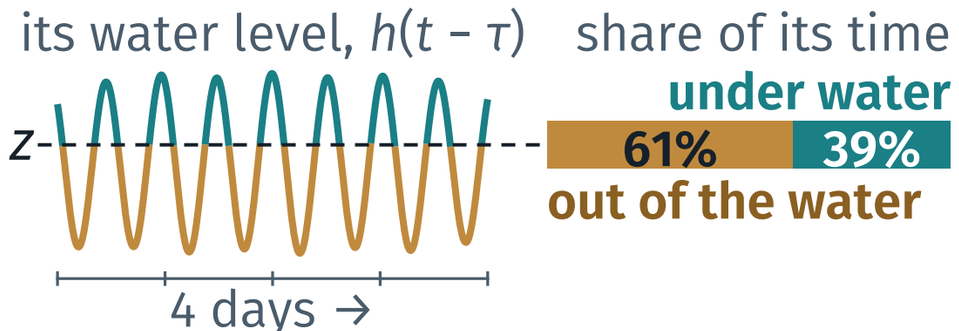


midpoint = elevation *z*

width = uncertainty of *z*

needs wet and dry looks, otherwise left empty

5 Its emerged time

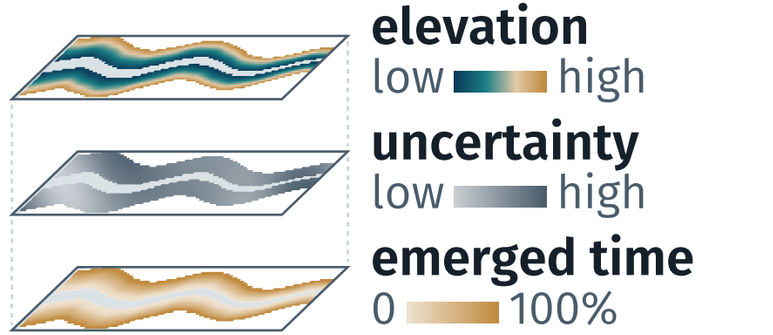


emerged time *E*(*z*) = share of time its water is below *z*: *h*(*t* − *τ*) < *z*

at all times, not only image times

OUTPUT

6 All pixels: the maps



base layer for the potential habitat of *Zostera noltei*

ecologists set the habitat limits